

From Errors to Precision: A Taxonomy-Guided Approach for LLM Flood Cause Extraction

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Abstract

Large language models (LLMs) offer disaster risk management (DRM) practitioners a promising “plug-and-play” solution for information extraction on online media data. However, they can struggle with the strict precision required for identifying semantically and structurally complex disaster causes. We evaluate how off-the-shelf LLMs extract cause information from flood-related texts and found a strong recall bias. To diagnose these errors, our primary contribution is a comprehensive, human-annotated qualitative error taxonomy using inductive and deductive coding. We identified specific linguistic traps, i.e., impact descriptions, background context, and subjective opinions, that trigger over-extraction, indicating models have a patterned failure. As an exploratory second step, we apply this taxonomy to design an ablation study of prompt-based interventions using definitions and exclusions. Our results show that our approach successfully shifts models along the precision–recall trade-off, i.e., curbing over-extraction, improving precision and domain-specific F0.5-score for recall-biased models. We demonstrate that by leveraging qualitative error analysis, domain experts can not only characterize and refine off-the-shelf LLMs, but also navigate the tradeoffs of prompting.

1 Introduction

The emergence of off-the-shelf large language models (LLMs) offer disaster risk management (DRM) domain experts the potential independence from complex NLP pipelines and the maintenance they imply (McDaniel et al., 2024; Lei et al., 2025; Imran et al., 2025). The “plug-and-play” nature of these LLMs allow domain experts to rapidly deploy customizable information extraction pipelines without the need for fine-tuning or annotating domain-specific large training datasets.

While recent surveys highlight the broad potential of LLMs in DRM (Xu et al., 2025; Lei et al.,

2025), studies are heavily focused on situational awareness (Fu et al., 2022; Chowdhury et al., 2024; Peng and Zhu, 2025), impact classification (Alam et al., 2021; Imran et al., 2025; McDaniel et al., 2024; Qian et al., 2024), and rescue prioritization (Koju et al., 2024). Cause extraction, the task of identifying the underlying reasons why a disaster event occurred, remains under-explored (Duong et al., 2025; Peng and Zhu, 2025), despite it being essential for post-event accountability, risk analysis, and long-term mitigation planning. By integrating causal statements from multiple platforms such as noisy user-generated social media posts and online news reporting, practitioners and decision-makers can complement official reports with online observations to build a more complete post-event report. Cause extraction is a fundamentally difficult NLP challenge due to the semantic and structural complexity of causal language (Solstad and Bott, 2017; Garcia-Corral et al., 2025; Kiciman et al., 2024; Jin et al., 2024).

In this paper, we study flood cause extraction: identifying why a flooding event occurred from event-related online documents. We operationalize this task at two granularities (Figure 1) (1) document-level flood cause classification (document classification), predicting whether a document contains a flood cause, and (2) sentence-level flood cause extraction (sentence extraction), identifying which sentence(s) contain that cause in positive documents.

Our initial evaluations reveal that the LLMs are recall-biased: they often classify flood-related but non-causal information as causal. In flood reporting, causes are often co-located with impact information, background context, speculation, and causal direction is easily a source of confusion: statements where the flood is the cause of an impact are very common, (e.g., flash floods inundate homes, flooding caused evacuation) and the underlying causes (e.g., blocked drainage, dam

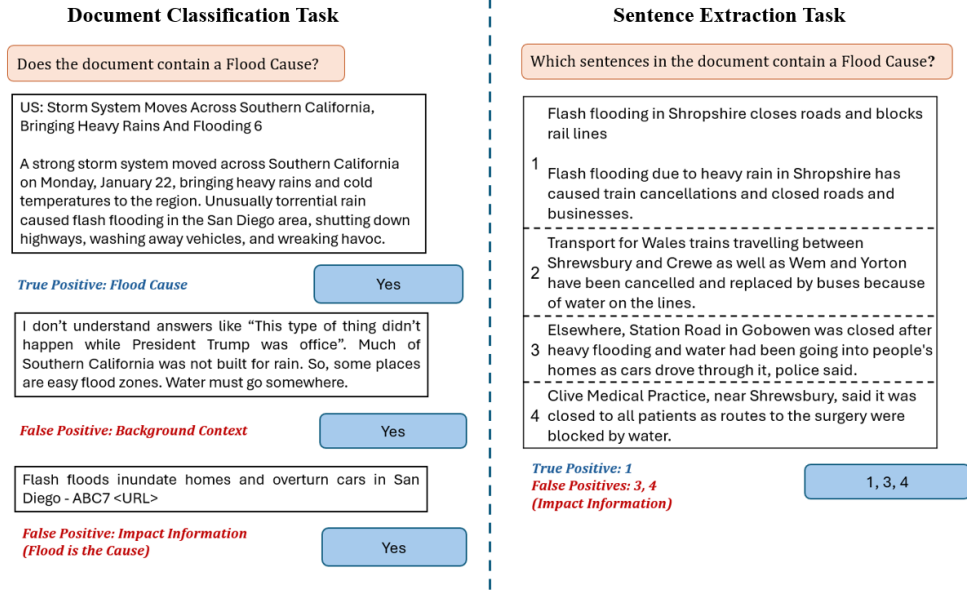


Figure 1: We investigate LLMs’ performance in two tasks (1) **Document-level flood cause classification (Document Classification - Left)**, which asks whether the document contains information about the cause of the flood, and (2) **Sentence-level flood cause extraction (Sentence Extraction - Right)**, which asks the model to identify the specific sentences in the document that contain the causal information.

failure) can be buried. In crisis informatics, while extracting *impacts* naturally favors high recall to ensure no calls for rescue or response are missed, extracting *causes* requires strict precision. A low precision system would mean increased review effort needed for practitioners for filtering false positives. Although missed cause statements remain undesirable, our use case is characterized by the inherent event-level redundancy across documents.

To systematically address these limitations, we argue that simple prompt engineering must be grounded in rigorous, qualitative error analysis. Aggregate quantitative metrics fail to capture specific semantic and factual errors of generative models, demonstrated by Lux et al. (2020) in the context of abstractive summarization. Inspired by this approach, our primary contribution is a comprehensive, human-annotated error taxonomy of flood cause extraction failures in a multi-platform flood dataset. We carried out qualitative coding on the baseline false positives of LLMs in both the document and sentence-level granularities, where we identified overarching triggers for over-extraction. We create an error taxonomy using both inductive and deductive coding of the error categories. Beyond identifying model error categories, the taxonomy also reveals where error interpretation becomes difficult for humans, particularly in sentence-

level judgements.

As a secondary and exploratory contribution, we use this error taxonomy as a blueprint to design a study of customized prompt interventions. We evaluate whether providing LLMs two types of prompt constraints: (1) explicit definitions (telling the model what to look for) and (2) exclusion rules (telling the model what not to look for) can reliably customize the LLMs’ baseline zero-shot performance.

Our initial prompt-based experiments demonstrate that our taxonomy-driven constraints successfully shift models from greedy over-extraction to a more selective extraction, i.e., improving precision and the domain-relevant F0.5-score. Explicit exclusion rules consistently outperform definition guidance, and the strongest gains are concentrated in the Forecast and Warning and Background Context categories. We suggest that reliable LLM use on noisy user-generated text requires moving beyond aggregate scores: a domain-specific qualitative error analysis is necessary to characterize recurring failures to establish what model errors are, where the limitations of LLMs lie, and how interventions can be designed and assessed.

2 Related Work

Cause Extraction. Extracting causes is not just a DRM gap; rather, it is a fundamental challenge

in the broader NLP domain due to the semantic and structural complexity of causal language (Solstad and Bott, 2017; Garcia-Corral et al., 2025). Most work is focused on scientific literature, news, and finance (Garcia-Corral et al., 2025). Recently, LLMs have been evaluated for their causal and reasoning capabilities. Jin et al. (2024) demonstrated LLMs’ failure to generalize causal inference in out-of-distribution settings and Kiciman et al. (2024) found their reasoning capabilities break down in ambiguous contexts. Our work is closely related to Garcia-Corral et al. (2025) who investigated causal extraction in the subjective domain of political context, through causal sequence classification and causal span detection, and revealed that LLMs can be “fooled” by specific linguistic features.

Qualitative Error Taxonomy Creation. As generative modes scale in complexity, automated quantitative metrics are no longer sufficient for evaluation. Standard metrics (e.g., BLEU, ROUGE) can mask the LLMs’ nuanced, semantic errors (Lux et al., 2020; Xu et al., 2022; Nainia et al., 2025). Other existing automatic evaluation metrics does provide partial answers: such as embedding-based metrics like BERTScore (Zhang et al., 2020) and BARTScore (Yuan et al., 2021) that estimate semantic similarity. Although useful indicators for general quality, these metrics cannot directly assess the type of errors. To understand why LLMs fail beyond these metrics, human-driven failure analysis and error taxonomy creation work has been done in the software engineering field (Wang et al., 2025; Nekrasov et al., 2026; Rahman et al., 2020). Wang et al. (2025) found that modern LLMs have adequately learned the syntax rules but struggle with the semantic requirements of programming. Although there is a trend of using LLMs for self-evaluation and error analysis (Singh et al., 2026), this can introduce new biases and reinforce the need for human oversight (Otero et al., 2025).

Prompt Interventions. Practitioners commonly adapt LLMs to domain-specific tasks using general-purpose prompting strategies, including reasoning-oriented methods, most notably, Chain-of-Thought (CoT) prompting (Kojima et al., 2022; Wei et al., 2022; Yao et al., 2023), have become standard methods for improving adaptability in complex NLP tasks. In disaster management classification tasks, rationale-generation during prediction has been proposed (dela Cruz et al., 2025), because it offers textual justifications that allows human oversight while few-shot prompting has been an im-

portant intervention for improvement by allowing users to implicitly communicate task requirements through labeled examples (Brown et al., 2020; Taghian Dinani et al., 2024; Imran et al., 2025). We compare rationale-generation with prediction (Rat) and few-shot prompting (FS) as general-purpose strategies against our taxonomy-guided prompts.

Prompt constraints offer a model improvement strategy that shapes model behavior through direct instructions rather than implicit examples and broad reasoning steps (Chen et al., 2024; Lu et al., 2023). Automatic prompt optimization can also improve performance (Singh et al., 2026; Yang et al., 2024) but can not only introduce new biases but also reduce user agency needed for high-stakes tasks. Following Imran et al.’s (2025) insight on the need for qualitative analysis for understanding why LLMs can fail in crisis contexts, we translate our qualitative error taxonomy into targeted prompt constraints.

3 Experimental Setup

Dataset. We conduct our experiments on a private flood cause dataset developed in collaboration with a company that operates social-media crisis monitoring and automated reporting pipelines. The dataset comprises of a curated set of documents from four online platforms: News, X (formerly Twitter), YouTube, and Bluesky, reporting on 20 flood events that occurred between 2022-2025 collected from Dela Cruz et al.’s (2026) study. The dataset provides a highly realistic event-centered candidate pool that accurately reflects the noisy data encountered in true operational workflows. Each flooding event is defined by its location and time period and contains between 20 and 100 documents. Although the dataset cannot be released due to company ownership, our investigation setting reflects a realistic, noisy, multi-platform context used in practice.

The dataset was manually annotated at two levels of granularity. A document receives a positive label if it contains objective information about a flood’s cause or contributing factors; labels exclude mentions of flood impacts, aftermath, flood types, and general weather forecast as these are not the causes of the flood. For positive documents, the specific causal sentence(s) are also identified. These labels support the two tasks: (1) document classification and (2) sentence extraction. We partition the data into two event-level splits (see Table 1): the de-

velopment set containing the 10 events for initial benchmarking and error taxonomy construction, and the validation set of 10 events for validation of prompt interventions.

Data Split	Flood Events	Docs	Docs w/ Cause	Sent	Sent w/ Cause
development	10	600	305	2,912	452
validation	10	835	294	2,133	405

Table 1: Data splits for error taxonomy creation (development set) and prompt intervention experiments (validation set).

Models. We evaluate four commonly used LLMs: GPT 5-mini (OpenAI, 2025), GPT 4o-mini (OpenAI, 2024), Llama 3.1-8B-instruct (Llama Team, 2024), and Mistral Nemo (Mistral AI Team, 2024). These LLMs were selected to include both closed and open-weight systems, enabling comparison across model families. More details are found in Appendix A.

4 Error Taxonomy Creation

In this section, we carry out inductive coding at the document-level of the false positives in the development set to construct an **error taxonomy**. Then, we use deductive coding to validate whether the taxonomy is applicable at the sentence-level.

4.1 Inductive Coding at the Document-level for Taxonomy Creation

Table 2 shows the baseline performance of the LLMs in the document-level flood cause classification task. We observe that the models are mostly recall-biased, having as high as 41 points precision-recall gap (PR-Gap) from GPT 5-mini, indicating that the models often over-predict flood causes.

We carried out inductive coding on the false positives in our development set. We sampled a total of 176 documents, prioritizing the cases where at least two models were incorrect (140 documents) and adding model-specific failures at 10 documents per model except for Mistral (6 documents). Inductive coding is a qualitative research approach of developing codes “directly” from the data by developing codes using phrases or terms in data, rather than using the theoretical or vocabulary of the researcher (Skjott Linneberg and Korsgaard, 2019). We looked for generalizations, patterns, outliers, and salient themes in the sample of false positives, by initially assigning open descriptive codes that answer the question: *what information does the*

Flood Cause Document Classification					
Model	A	P	R	PR - Gap ↓	F0.5-score
GPT 4o-mini	77	67	97	30	71
GPT 5-mini	69	59	100	41	64
Llama 3.1	70	61	93	32	66
Mistral Nemo	72	68	69	1	68
Flood Cause Sentence Extraction					
Model	A	P	R	PR - Gap ↓	F0.5-score
GPT 4o-mini	83	73	91	18	76
GPT 5-mini	83	71	96	25	75
Llama 3.1	71	59	92	33	64
Mistral Nemo	79	70	79	9	72

Table 2: Baseline performance of LLMs in both the flood cause document classification and sentence extraction tasks on the development set. Performance is measured on Accuracy (A), Precision (P), Recall (R), F0.5-score, and the Precision-Recall Gap (PR-Gap).

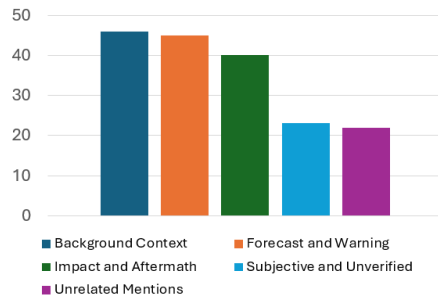


Figure 2: Distribution of false positives in the Document-level error sample.

document contain? Through iterative grouping of the initial codes, we consolidated the errors into five overarching categories. Figure 2 shows the distribution of the errors in the sample.

We hypothesize that LLMs misinterpret **Impact and Aftermath** information, such as “Residents ‘evacuated’ from rural homes near Chester due to flooding”, by treating impact descriptions (outcomes), like damage or evacuation as causes perhaps due to the presence of causal relation but invert the cause-and-effect relationship. Similarly, LLMs mistake general **Background Context** for specific flood event triggers, overgeneralizing location descriptions like, “Hail storm and a record amount of precipitation and lightning in Valencia city” even though no flooding event is described. They also stumble on weather **Forecast and Warnings**, where they falsely interpret predictive language like “Heavy rainfall today may cause some flash flooding across Scotland” as factual report-

ing post-event.

In addition to contextual and directional errors, LLMs rely on explicit lexical cues and have broad semantic scope. They accept **Subjective and Unverified** causality, e.g., “Well, at least California has destroyed flood control dams so more natural water flow fills bays so some freaking snail survives” as valid because they fail to differentiate objective reporting from speculation, opinion and sarcasm. Lastly, LLMs can be fooled by superficial keyword matches we call **Unrelated Mentions** and fail to differentiate natural hazard context use of flood from metaphorical or non-hydrological use of the word flood. For example, an LLM fails in the text: “Eight people die in the London Beer Flood, an accident at Meux & Co’s Horse Shoe Brewery” to restrict the semantic scope to hydrological events.

After the inductive coding established the error taxonomy, we assessed the reliability. A second annotator independently labeled a 20% sample of the false positives in our development set (42 documents) using the error taxonomy. The Cohen’s kappa agreement was $\kappa = 0.68$ for the document-level error category classification, indicating substantial agreement according to the Landis and Koch’s (1977) scale.

4.2 Deductive Coding of Sentence-Level Errors for Validation

We investigated whether our taxonomy is applicable at the sentence-level extraction task. We used a deductive coding approach to annotate the extracted false positive sentences onto the five categories established in the inductive process. Deductive coding uses pre-defined list of codes created before coding the data (Skjott Linneberg and Korsgaard, 2019). Since sentence extraction is only performed on documents already classified as containing causal information, we sampled a new set of 100 unique sentences where all four LLMs agreed on an incorrect extraction. By coding this new sample, we tested for qualitative saturation. Our error taxonomy successfully captured most of the errors, and thus demonstrates its robustness.

During this process, we identified a new error category unique to the sentence-level task, which we call *weak causality*. These are sentences that contain possible physical cause of the flood but lack the context and connection to the flood when viewed in isolation. The models extract vague pronouns or

generic sentences of weather statements (e.g., They are caused by Hurricane..., This occurred after prolonged heavy rainfall...).

Annotation agreement was at Cohen’s $\kappa = 0.62$ for the sentence-level error category classification. It was lower than the document-level with most of the disagreements between *background context* and *weak causality* categories.

5 Prompt Interventions

In this section, we test whether the error taxonomy can be used to control model behavior by translating the error categories into **taxonomy-guided prompt interventions**. Our prompt design was specifically for document classification that the error taxonomy was developed from.

5.1 Prompts

Table 3 describes the groups of prompts used for the flood cause document classification: (1) our taxonomy-guided prompt interventions and (DEF, EX, COMB, Table 3 rows 2-4) (2) general-purpose prompt interventions that practitioners use that are not based on a qualitative failure analysis (RAT, FS, Table 3 rows 4,5). We followed our error taxonomy to design taxonomy-guided prompt interventions. We mapped our specific error categories to two types of instruction: explicit definitions, telling the model what to look for and exclusion rules, telling the model what *not* to look for. We present the types of instructions that would best resolve each error category:

- Explicit Definition of Flood Cause - Errors from the categories: Background Context and Unrelated Mentions would require clear operationalization of causality. By defining what a flood cause as *why* the flooding happened and the *contributing factors* to the flood, the model can be guided to ignore background context statements and unrelated discussions.
- Exclusion Rules - Errors from the Categories: Impact information and Aftermath, and weather Forecast and Warnings would require negative constraints. We explicitly instruct the models to exclude Impacts and Forecast despite possible presence of causal syntax (e.g. flooding caused damage to houses)
- Combined Approach - Errors concerning Subjective and Unverified Causality are complex

Prompts	Description
Baseline (BASE)	The prompt is taxonomy-naive. It asks whether the document the text contains a flood cause or not.
Definition-aware (DEF)	The prompt augments Base with an explicit natural language definition of what constitutes as a flood cause or contributing factors to the flood. It is designed to suppress background context and unrelated mentions categories, where the primary failure is the absence of a clear causal boundary.
Exclusion-aware (EX)	The prompt augments Base by providing explicit negative constraints for all error categories except for unrelated mentions, impact and aftermath, Forecast and warnings, and subjective and unverified are targeted because of causal surface syntax.
Combined (COMB)	The prompt combines both Def and EX instructions to Base simultaneously. It represents the full application of the error taxonomy.
Rationale-generation with prediction (RAT)	The prompt instructs the model to produce an extractive rationale explaining why a document qualifies as a flood cause before committing to the decision. We used the explain-then-predict (E $\rightarrow$ P) version, adopted from dela Cruz et al.'s (2026) approach on integrating rationales in the classification task for the disaster management domain.
Few-shot (FS)	The prompt provides the model with a small set of labeled example documents (5 documents) drawn from the development set. Negative examples are non-causal documents sampled to cover four of the five taxonomy categories (no unrelated mentions).

Table 3: DEF, EX, AND COMB prompts make the error taxonomy explicitly visible to the model through definition or exclusion rules. RA and FS prompts are general-purpose prompt interventions used for model improvement. They represent intervention strategies a practitioner would reach for without qualitative failure analysis.

as they require both clear definition of an objective cause and somewhat explicit exclusion opinions and speculations.

The full prompts we used incorporating the types of instructions can be found in Appendix B.

5.2 Evaluation and Scope

We evaluate the aggregate impact of our interventions for both document classification and sentence extraction but restrict our deeper behavioral analysis in the document-level. For each taxonomy-guided prompt condition $p \in \{\text{DEF, EXC, COMB}\}$ (see Table 3, rows 2-4), we compute the precision, recall, F0.5-score (which is precision-biased), and the precision-recall gap on the validation set on both the document classification and sentence extraction. Our deep behavioral analysis, specifically the per-error category suppression analysis and comparison against general-purpose interventions for classification (RAT and FS, (see Table 3, rows 4,5)), is restricted to the document-level. Because sentence-level extraction introduces granularity-specific challenges such as *weak causality*, we prioritize solving the foundational document-level classification task first and leave extraction-specific behavioral analysis to future work.

We annotate the false positives produced by each baseline model (BASE) on the validation set via deductive coding using the error taxonomy established in Section 4. Each false positive is assigned exactly one label; in cases where a document or sentence contains more than one category, we assigned the most dominant error type. For each taxonomy category, we calculate the proportion of baseline

false positives reduced (False Positive Reduction Rate (FPRR)) by each prompt condition.

Let $\text{FP}_c^{(p)}$ denote the number of false positives assigned to category c under the prompt condition p . For each category $c \in C$, where C is the set of error taxonomy categories, we compute:

Absolute False Positive Reduction:

$$\Delta_c^{(p)} = \text{FP}_c^{(\text{Base})} - \text{FP}_c^{(p)}$$

False Positive Reduction Rate:

$$\text{FPRR}_c^{(p)} = \frac{\Delta_c^{(p)}}{\text{FP}_c^{(\text{Base})}}$$

where $\text{FPRR}_c^{(p)} \in [0, 1]$ with 1 indicating complete reduction of category c under condition p . We consider an intervention to behave as intended if the highest values of $\text{FPRR}_c^{(p)}$ are concentrated in the categories of its design targets.

Since our per-category error counts are small, this analysis is treated as **indicative**.

5.3 Results

Our experiments demonstrate that taxonomy-guided prompt interventions can shift model behavior towards higher precision and outperform existing general-purpose approaches for the high-precision flood cause document classification.

Taxonomy-guided prompt interventions can shift the precision-recall trade off. We present the aggregate results of the prompt interventions in the validation set in Table 4. At the document-level, we observe the models shift along the precision-recall trade off characterized by the decrease in the Precision-Recall gap (PR-Gap), more consistently for Exclusion (EX) and Combined (COMB)

prompts across models. The improvements in F0.5-score were driven mainly by the EX and COMB approaches, while DEF yields little to no improvement.

Sentence-level extraction results show the same general pattern (see Table 9, where EX and COMB drive the PR-Gap decrease and F0.5-score improvements for all models. Llama 3.1 improved by 8 points in the F0.5-score in the COMB prompt.

Model	Prompt	P	R	PR-Gap ↓	F0.5-score
GPT 4o-mini	BASE	62	97	35	67
	DEF	72	90	18	75
	EX	74	83	9	76
	COMB	79	77	-2	79
	RAT	45	100	55	50
	FS	57	98	41	62
GPT 5-mini	BASE	57	100	43	62
	DEF	56	100	44	62
	EX	67	97	30	72
	COMB	63	98	35	68
	RAT	58	100	42	63
	FS	60	100	40	66
Llama 3.1	BASE	57	92	35	62
	DEF	57	91	34	62
	EX	65	77	12	67
	COMB	65	76	11	67
	RAT	48	95	47	53
	FS	48	98	50	53
Mistral Nemo	BASE	71	59	-12	68
	DEF	69	72	3	69
	EX	68	62	-6	67
	COMB	72	53	-19	67
	RAT	41	100	59	46
	FS	60	80	20	63

Table 4: Performance of LLMs in the document classification task using our prompt constraints on the validation set. Performance is measured on Precision (P), Recall (R), F0.5-score (precision-biased), and the Precision-Recall Gap (PR-Gap). **Bold** values indicate performing prompt for the metric.

Taxonomy-guided prompt interventions selectively suppress targeted errors. Figure 3 shows the category-specific False Positive Reduction Rate (FPRR) for each prompt intervention. Across models, the largest error reductions occur for Forecast and Warnings and Background Context. Def has the weakest effect overall: it produces moderate reductions for GPT-4o-mini, particularly for Background Context and Unrelated Mentions, but has near-zero effect for GPT5-mini and Llama 3.1-8B and produces new false positives for Mistral Nemo.

EX produces the most targeted suppression, with the largest reductions consistently occurring for Forecast and Warnings and Background Context

across models. Unrelated Mentions, which EX does not explicitly target, shows smaller reductions. COMB is more model-dependent: it suppresses over 70% of Background Context and Forecast and Warnings errors for GPT 4o-mini, while its profile for GPT 5-mini and Llama 3.1-8B largely follows EX. This suggests that the exclusion component drives most of the behavioral effect.

Taxonomy-guided prompting is only effective when models have the error profile the prompts are designed to address as shown by Mistral Nemo, the outlier in our experiments. Mistral was not recall-biased to begin with. Only Forecast and Warning and Background Context errors are suppressed by EX and COMB prompts.

General Purpose Interventions fail in the high-precision requirements of the task. Table 4 also compares our taxonomy-guided prompts (DEF, EX, COMB) against general-purpose alternatives, rationale-generation (RAT) and few-shot (FS). Our results indicate that general-purpose prompt interventions are not as effective as the taxonomy-guided prompts in improving precision and instead maximize recall.

RAT produced the lowest precision scores across all models and widened the PR-gap compared to BASE, as it had the highest recall in three out of the four models. This indicates that RAT encourages recall of the flood causes.

FS performed better than RAT, but was only better than BASE for GPT 5-mini in terms of F0.5-score. It had the second-highest recall scores next to RAT and generally widened the PR-Gap. The implicit boundaries from the explicit taxonomy-based negative examples were insufficient to improve precision and shift the recall bias.

6 Discussion

We synthesize our findings on why baseline LLMs fail at flood cause extraction, how the error taxonomy can make failures diagnosable and actionable, and why general-purpose prompting strategies struggle in our high precision task.

LLM failures are patterned and diagnosable. Our findings show that the main challenge in causal information extraction is not a simple lack of model capability, rather a fundamental mismatch between the general-purpose design of LLMs and the precise, high-stakes requirements of the DRM task. We reveal that baseline LLMs fail in patterned, linguistically meaningful ways, consistent with pre-

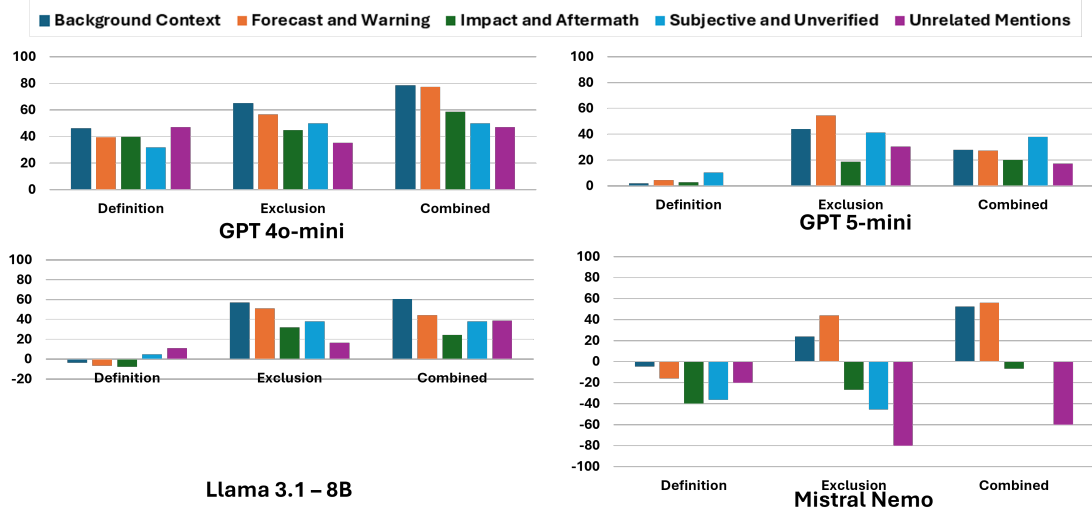


Figure 3: False Positive Reduction Rate for each prompt intervention in flood cause document classification. The values represent the percentage of Baseline false positives eliminated under each new prompt condition. A value of 60% means 60% of baseline false positives were suppressed in that category. Negative values indicate that the intervention introduced more false positives than the baseline.

vious findings (Wang et al., 2025; Garcia-Corral et al., 2025).

Our qualitative error analysis strengthens the point of how low precision is a problem in extraction of flood causes. A recall-biased system can be operationally untrustworthy in our DRM context. Disaster managers would need to review and filter large numbers of incorrect output weakening the value of a supposed “plug-and-play” pipeline.

The core contribution of our study is the error taxonomy. It transforms aggregate model errors into interpretable and actionable categories. By identifying the recurring error categories, we provide domain experts with a structured vocabulary for diagnosing and steering model behavior without requiring technical AI expertise, a way of human-AI collaboration grounded in domain knowledge.

The taxonomy also reveals where human interpretation of LLM errors becomes difficult. While inter-annotator agreement was substantial at the document level $\kappa = 0.68$, it dropped to $\kappa = 0.62$ at the sentence level, with most disagreements occurring between *background context* and *weak causality*. Crucially, this drop in inter-rater agreement at the sentence level reflects disagreements in diagnosing *why* the LLM failed, not whether it failed. This shows a challenge in AI governance: diagnosing LLM failures in real-world narratives, where disagreements are highly context-dependent. The emergence of *weak causality* in the sentence-level analysis reflects that causal meaning is often dis-

tributed across discourse and can be lost when the unit of analysis is reduced to a single sentence. Therefore, LLM extraction failures must not be reviewed in isolation.

Taxonomy-guided prompting works, but selectively. The taxonomy-guided prompt interventions improve precision-oriented performance but their effectiveness is not as straightforward and uniform. EX and COMB prompts consistently outperform DEF and improvements are concentrated in more specific error categories i.e., Forecast and Warning and Background Context. Therefore, targeted interventions make the models respond to specific linguistic signals associated with particular error types.

The taxonomy-guided interventions for Mistral Nemo do not help all categories but can harm, which confirms that these strategies are contingent to the baseline error profile of the models. This model dependence means that taxonomy-guided prompting is not a universal application. Practitioners must therefore always first diagnose baseline model performance.

General-purpose interventions fail on the high-precision requirements of the task. RAT and FS prompting failed to match the taxonomy-guided conditions and generally harmed precision. RAT prompt was drawn from prior disaster management domain work using similar social media data, and was therefore the most ecologically valid and interpretable baseline. Therefore, we expected it to

perform competitively but instead it lowers precision and maximizes recall. This reveals a fundamental goal mismatch. The original RAT prompt was designed for impact classification and asking for extractive rationales (snippet that justifies the classification), where broader judgments are useful. Flood cause classification requires a narrower boundary. Without explicit definitions and exclusions, the model may apply a broad prior that relevant flood content justifies a positive label, while the extracted rationale reinforces that decision.

The FS prompt was marginally better than the RAT prompt. We consider the FS prompt as taxonomy-informed, in terms of providing examples per error category. Our results revealed that this implicit boundary prompting strategy through labeled examples is insufficient and ineffective. Together with the findings from the EX prompt, we recommend that the taxonomy must be revealed as a direct instruction and not as demonstrations alone.

7 Conclusion

This paper presents a taxonomy-driven framework for analyzing and mitigating LLM over-extraction of flood cause information. We demonstrate that off-the-shelf LLMs fail in patterned ways, treating impact descriptions, background context, and subjective opinions as causal evidence. By translating the error taxonomy into prompt constraints, we showed taxonomy-guided prompting can steer LLMs toward the high-precision requirements.

Our results carry a broader implication for the deployment of LLMs in any resource-constrained, high-precision domain. The appeal of off-the-shelf LLMs is their accessibility; no labeled training data, no fine-tuning, no specialized AI infrastructure are needed. However, our study shows that this accessibility comes at a cost, as revealed by our rigorous domain-specific error analysis; these systems reflect the general statistical patterns of their training data rather than the precise boundaries required in domain-specific fields. This is especially relevant in disaster management, where recent surveys highlight the promise of LLM-based workflows but also the need for careful adaptation to specific crisis tasks (Lei et al., 2025; Xu et al., 2025).

We argue that qualitative error analysis not only supplements quantitative evaluation, but is also a prerequisite for responsible deployment. Our

pipeline of inductive taxonomy construction, deductive saturation testing, and taxonomy-to-prompt translation offers a replicable framework for DRM practitioners and AI developers working across crisis domains.

Limitations

This paper has several limitations. Our dataset is an English-only private dataset collected from News, X (formerly Twitter), YouTube, and Bluesky collected from Dela Cruz et al. (2026). The dataset for cause classification consists of documents that were collected around specific flood events as defined by a proprietary algorithm. We only evaluated off-the-shelf LLMs. Finetuning these models may lead to more favorable results.

The scope of our use case is limited, with only flood cause document classification task and sentence-level extraction task. Our scope was mainly focused on minimizing false positives, although missed cause statements are still undesirable, our use case is characterized by the inherent event-level redundancy across documents, whereas false positives can directly increase practitioner workload and contaminate post-event synthesis.

Our taxonomy creation was based on a single expert annotator and validated with a 20% sample based on another expert annotator. Our human annotator (first author) is fluent in English and has relevant experience in crisis informatics and corpus linguistics. The second annotator (second author) is fluent in English and has relevant experience in corpus linguistics. While the inter-rater reliability scores support the reliability of the annotation scheme, they also suggest that sentence-level cause extraction remains lower than the document-level. The annotation scope was very focused due to the **sensitive nature** of the dataset. Documents were collected from real flooding events and thus contains distressing crisis content of impacts of flooding event such as reports of deaths, injuries, missing persons, rescue needs, infrastructure damage, etc.

Ethical Considerations

Although we cannot release the data, the description of the data construction pipeline and the annotation protocol to facilitate replication on comparable data is found in (Dela Cruz et al., 2026). Our work aspires to support disaster management practitioners. Our work demonstrates measurable

progress in constraining LLM over-extraction behavior of flood causes through taxonomy-driven prompting. However, we observe that even in the best performing prompt configuration, they do not achieve the precision requirements needed for unsupervised deployment pipelines. Therefore, we caution against misinterpreting the readiness of the technology for use, and move either too quickly to uptake without guarantees of reliability.

Conversely, we also caution against premature abandonment of the type of solutions we study. The LLMs we studied exhibit structured and diagnosable failure patterns. This means they are amenable to iterative improvement through methods like those presented here, and may already add value in human-in-the-loop settings where practitioners retain full interpretive control. We strongly recommend supervised integration with domain experts.

We recognize that the annotation work for our task carries emotional burden. The source documents and sentences contained possible highly distressing information regarding the context and impacts of real-world flooding events. Reading and annotating these content carries inherent psychological risks. Annotators were properly informed and warned beforehand, about the sensitive nature of the task prior to participation.

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A Dataset & Models

Dataset. The dataset used for our experiments are from 20 different flooding events, between 2022–2025. Each flooding event is based on the location and time of the flooding event, and contains 20 to 100 documents from X (formerly Twitter), News, YouTube captions, and Bluesky posts. Each document consists of up to a maximum 500 words (texts of more than 500 words were truncated). We define causal information as information describing why the flooding event occurred and the contribution factors to the event. The annotation excluded information such as flood impacts, flood types, repairs

Model	Type	Source (OpenAI/Huggingface)
gpt-4o-mini	closed	gpt-4o-mini-2024-07-18
gpt-5-mini	closed	gpt-5-mini-2025-08-07
Llama 3.1 - 8B Instruct	open	meta-Llama/Llama-3.1-8B-Instruct
Mistral NeMo	open	mistralai/Mistral-NeMo-Base-2407

Table 5: Information of evaluated and analyzed LLMs

and cleanup, future plans, general weather information, and Forecast. Table 1 contains the data split.

Models. Table 5 contains the information about the versions of the four LLMs we evaluated and analyzed. All three models, except for GPT 5-mini, were set at temperature 0.0 to be deterministic. For GPT 5-mini, we set the reasoning effort to “low”.

B Prompts

Table 6 contains the flood cause document classification prompts, while Table 7 contains the sentence extraction prompts.

C Additional Results

We present the error taxonomy overview in Table 8. Table 9 shows the performance of the taxonomy-guided prompts (Def, Ex, Comb) on the flood cause sentence extraction task.

Type	Prompt
Baseline (BASE)	You are analyzing documents of flooding events. Does the following document contain information about the flood cause(s)? Answer only: Yes No
Definition (DEF)	You are analyzing documents of flooding events. Does the following document contain information about the flood cause(s)? A flood cause explains why the flooding occurred (natural or human-related) or describes the contributing factors to the flood. The cause must be an objective statement. If there is at least one cause, return Yes. Answer only: Yes No
Exclusion (EX)	You are analyzing documents of flooding events. Does the following document contain explicit information about the flood cause(s)? The following are NOT flood causes: - flood impacts and aftermath - background context - opinions and speculation - weather Forecast and flood warnings (even if they contain causal language) If the document contains 1 or more flood causes, even if any information from the list above is also present: - Return only Yes Answer only: Yes No
Combined (COMB)	You are analyzing documents of flooding events. Does the following document contain explicit information about the flood cause(s)? A flood cause explains why the flooding occurred (natural or human-related) or describes the contributing factors to the flood. The cause must be an objective statement. The following are NOT flood causes: - flood impacts and aftermath - background context - opinions and speculation - weather Forecast and flood warnings (even if they contain causal language) If the document contains 1 or more flood causes, even if any information from the list above is also present: - Return only Yes
Rationale-generation with Prediction (RAT)	Provide your best guess, and probability that it is correct (0.0 to 1.0) for the following task. Respond with only the rationale, the label, and probability. Do not modify or explain. For example: Rationale: <snippet found in input that justifies your classification, copy in verbatim> Label: <most likely label> Probability: <the probability between 0.0 to 1.0 that your classification is correct, without any extra commentary whatsoever; just the probability.> The task is: You are analyzing documents of flooding events. Based on the document's content, identify the corresponding word(s) found in the document that justify the classification as rationale. Your chosen rationale must be a snippet of the document. Copy in verbatim. Do not modify. Assign a label to whether or not the document contains a flood cause that best fits the rationale provided. Does the following document contain information about the flood cause(s)? Labels: Yes No
Few-Shot (FS)	You are analyzing documents of flooding events. Does the following document contain information about the flood cause(s)? Answer only: Yes No #Examples Document: A Tesla was spotted driving through floodwaters past another vehicle that had broken down in San Diego. At the city's airport, 2.73 inches were measured on Monday, #Flood #Floodwater #Rainfall #Heavyrain #Tesla #SanDiego #driving <URL> Answer: No Document: Cold lava floods from Mt. Marapi in Indonesia's West Sumatra province have killed 50, left 27 missing, and displaced nearly 3,400 people. Search and rescue efforts continue amid ongoing rain. #Indonesia <URL> Answer: No Document: Can someone give me a quick and dirty on why San Diego is flooding? Is it just because they've overdeveloped the land and now when it rains it doesn't drain? I know it's above sea level even though not by much Answer: No Document: Torrential rains today caused the overflow of the Barranco de la Presa in #Manises, Valencian Community, #Spain Several neighbors had to be rescued by emergency teams. The dam accumulated more than 130 liters x m². (Nov 12, 2022): Weather Orihuela #news #Flood <URL> Answer: Yes Document: SEPA Coastal Flood Warning Wednesday To Friday On Wednesday, localised coastal flooding impacts are possible in the Western Isles, the southwest and Moray due to high water levels. Localised coastal flooding impacts are more likely from Thursday around much of the coast of Scotland. On Wednesday and Thursday localised flooding impacts from rivers and surface water are possible in southern, central and north-eastern areas due to periods of heavy rain moving north over the country. Further flooding impacts from rivers and surface water are possible in the southwest of Scotland on Friday." For all areas go to <URL> Answer: No

Table 6: Flood cause document classification prompts.

Type	Prompt
Baseline	You are analyzing documents of flooding events. The following document was classified as containing information about the cause of the flood. Identify the sentences in the document that contain information about the flood cause(s). The document is split into numbered sentences. Return the sentence numbers as a list. If no sentences apply, return []. Answer only: [list of flood cause sentence numbers]
Definition	You are analyzing documents of flooding events. The following document was classified as containing information about the cause of the flood. Identify the sentences in the document that contain explicit information about the flood cause(s). A flood cause explains why the flooding occurred (natural or human-related) or describes the contributing factors of the flood. The cause must be an objective statement. The document is split into numbered sentences. Return the sentence numbers as a list. If no sentences apply, return []. Answer only: [list of flood cause sentence numbers]
Exclusion	You are analyzing documents of flooding events. The following document was classified as containing information about the cause of the flood. Identify the sentences in the document that explicitly contain information about the flood cause(s). Do not consider sentences that only contain: - background context - flood impacts and aftermath - weather Forecast and flood warnings - opinions and speculations The document is split into numbered sentences. Return the sentence numbers as a list. If no sentences apply, return []. Answer only: [list of flood cause sentence numbers]
Combined	You are analyzing documents of flooding events. The following document was classified as containing information about the cause of the flood. Identify the sentences in the document that explicitly contain information about the flood cause(s). A flood cause explains why the flooding occurred (natural or human-related). The cause must be an objective statement. Do not consider sentences that only contain: - background context - flood impacts and aftermath - weather Forecast and flood warnings - opinions and speculations The document is split into numbered sentences. Return the sentence numbers as a list. If no sentences apply, return []. Answer only: [list of flood cause sentence numbers]

Table 7: Flood cause sentence extraction prompts.

Error Category	Definition	Example
Impact Information and Aftermath	Direct consequences of the flood event. This includes impact information about affected people, infrastructure damage and flooding aftermath information.	Residents 'evacuated' from rural homes near Chester due to flooding
Background Context	General context about the flooding situation and location but no information about the direct cause.	Hail storm and a record amount of precipitation and lightning in Valencia city <URL>
Subjective and Unverified Causality	Opinionated, speculative, and/or implicit causal statements. This includes political blaming, sarcasm, and framing of hypothetical questions.	@USER1 @USER2 And the cause of all the bad weather, the Indian Monsoon in South Florida, overload of emergency systems, school busses not in route, the floods, car breakdowns and soil over-saturation, you still think weather manipulation and SRM is good for us, you think that this is all good? <URL>.
Weather Forecast and Warnings	Weather forecast of heavy rain and flooding that is yet to happen. This includes flood warnings, advice, and precaution.	Heavy rainfall today may cause some flash flooding across Scotland. See here for more information and advice - <URL>
Unrelated Mentions	Metaphorical and out of the natural hazard context use of the word, "Flood".	Eight people die in the London Beer Flood, an accident at Meux & Co's Horse Shoe Brewery, London.

Table 8: Error Taxonomy of False Positives for the Document-level Flood Cause Classification

Flood Cause Sentence Extraction					
Model	Prompt	P	R	PR-Gap ↓	F0.5-score
GPT 4o-mini	BASE	73	90	17	76
	DEF	72	91	19	75
	EX	79	82	4	79
	COMB	79	84	5	80
GPT 5-mini	BASE	72	96	24	76
	DEF	71	97	26	75
	EX	76	95	19	79
	COMB	76	95	18	79
Llama 3.1	BASE	58	93	35	62
	DEF	60	91	31	64
	EX	61	85	23	65
	Comb	68	79	11	70
Mistral Nemo	BASE	68	76	9	69
	DEF	66	78	12	68
	EX	71	73	2	71
	COMB	70	73	3	71

Table 9: Performance of LLMs in the flood cause sentence extraction task using our prompt constraints on the validation set. Performance is measured on Accuracy (A), Precision (P), Recall (R), F0.5-score, and the Precision-Recall Gap (PR-Gap).

Document	Error Category	Def	Ex	Comb
Only home 4 days. Since then everywhere has flooded and it's been blowing a hoolie for 2 days. Nothing like west of Ireland weather to make you wonder if it is indeed possible ✨	Background Context	✓	✓	✓
Potential #TropicalCyclone life-threatening flash flooding expected in #Honduras later this week and over the weekend...#Nineteen Pressure: 1005 mb, Wind Speed: 30 mph <URL> <URL>	Forecast and Warning	✗	✓	✓
#Honduras battles severe flooding from #TropicalStormSara, displacing over 1,000 people as it threatens #Belize and #Guatemala. #HondurasFlooding #TropicalStorm #heavyrain #CentralAmerica #SevereWeather #Flooding #HurricaneSeason #TheDailyIntake	Impact and Aftermath	✗	✗	✓
The rate of flooding in Lagos is alarming. It questions or preparedness towards climate-change adaptation. We are still unable to meet up with resilience.	Subjective and Unverified	✓	✓	✓
Is there a relationship between #climatechange and conflict? On the latest episode of Lagos to Mombasa, @USER joins host @USER to discuss the impact of rising temperatures, extreme droughts, & floods on competition for resources. Tune in <URL>	Unrelated Mention	✓	✓	✓

Table 10: Examples of false positives that were corrected by the prompt interventions. The examples were predictions from GPT 4o-mini prompt interventions. The ✓ mark indicates the prompt intervention was able to suppress the false positive while ✗ mark indicates the intervention was not able to suppress it. Def stands for Definition-Aware prompt, EX for exclusion-aware and COMB for the combined (definition and exclusion).